

Pakistan Navy Engineering College, Karachi
National University of Sciences & Technology, Pakistan





EE 869 – Linear System Theory

Fall 2026

Lectures 1 – 3

State models

We will deal with dynamical systems modeled by n couple first-order ordinary differential equations:

$$\begin{aligned}
\dot{x}_1 &= f_1(t, x_1, \ldots, x_n, u_1, \ldots, u_m) \\
\dot{x}_2 &= f_2(t, x_1, \ldots, x_n, u_1, \ldots, u_m) \\
&\vdots \qquad\quad \vdots \\
\dot{x}_n &= f_n(t, x_1, \ldots, x_n, u_1, \ldots, u_m)
\end{aligned}$$

$$\dot{x}_i = \frac{dx_i}{dt}, \quad t \text{ is the time variable}$$

$u_1, \ldots, u_m$ are the input variables

$x_1, \ldots, x_n$ are the state variables

$$
\begin{aligned}
\dot{x}_1 &= f_1(t, x_1, \ldots, x_n, u_1, \ldots, u_m) \\
\dot{x}_2 &= f_2(t, x_1, \ldots, x_n, u_1, \ldots, u_m) \\
&\vdots \quad\quad \vdots \\
\dot{x}_n &= f_n(t, x_1, \ldots, x_n, u_1, \ldots, u_m)
\end{aligned}
$$

Vector Notation:

$$
x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ \vdots \\ x_n \end{bmatrix}, \quad u = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_m \end{bmatrix}, \quad f(t, x, u) = \begin{bmatrix} f_1(t, x, u) \\ f_2(t, x, u) \\ \vdots \\ \vdots \\ f_n(t, x, u) \end{bmatrix}
$$

$$
\dot{x} = f(t, x, u)
$$

Output equation:

$$y = h(t, x, u)$$

$$y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_p \end{bmatrix}, \quad h(t, x, u) = \begin{bmatrix} h_1(t, x, u) \\ h_2(t, x, u) \\ \vdots \\ h_p(t, x, u) \end{bmatrix}$$

State Model:

$$\begin{aligned} \dot{x} &= f(t, x, u) \quad \text{State equation} \\ y &= h(t, x, u) \quad \text{Output equation} \end{aligned}$$

Example: Consider a single-input–single-output (SISO) system represented by the nth-order differential equation

$$y^{(n)} = g\left(t, y, y^{(1)}, \ldots, y^{(n-1)}, u\right)$$

where u is the input, y is the output, and $y^{(i)} = \dfrac{d^i y}{dt^i}$

State variables:

$$x_1 = y, \quad x_2 = y^{(1)}, \quad \cdots, \quad x_n = y^{(n-1)}$$

$$\begin{aligned}
\dot{x}_1 &= x_2 \\
\dot{x}_2 &= x_3 \\
&\vdots \quad\quad \vdots \\
\dot{x}_{n-1} &= x_n \\
\dot{x}_n &= g(t, x_1, x_2, \ldots, x_n, u)
\end{aligned}$$

Classification of State models

Time-varying system:

$$\begin{aligned} \dot{x} &= f(t, x, u) \\ y &= h(t, x, u) \end{aligned}$$

Time-invariant system:

$$\begin{aligned} \dot{x} &= f(x, u) \\ y &= h(x, u) \end{aligned}$$

The response of a time-invariant system is invariant to shifts of the time origin. Therefore, without loss of generality, we can take the initial time to be $t_0 = 0$

Classification of State models

Nonlinear system:

$$\begin{aligned} \dot{x} &= f(t, x, u) \\ y &= h(t, x, u) \end{aligned}$$

Linear system:

$$\begin{aligned} \dot{x} &= A(t)x + B(t)u \\ y &= C(t)x + D(t)u \end{aligned}$$

A is $n \times n$, B is $n \times m$, C is $p \times n$, D is $p \times m$

The response of a linear system satisfies the superposition principle:

Let
$(x_a(t), y_a(t))$ be the response to $x_a(t_0) = x_0^a$ & $u_a(t)$
$(x_b(t), y_b(t))$ be the response to $x_b(t_0) = x_0^b$ & $u_b(t)$
then, for any real numbers α and β, $\alpha x_a(t) + \beta x_b(t)$ is the response to $x(t_0) = \alpha x_0^a + \beta x_0^b$ & $u(t) = \alpha u_a(t) + \beta u_b(t)$

Example: The Pendulum equation (with u = 0)

$$\begin{aligned} \dot{x}_1 &= x_2 \\ \dot{x}_2 &= -(g/\ell)\sin x_1 - (k/m)x_2 \\ y &= x_1 \end{aligned}$$

$$f = \begin{bmatrix} x_2 \\ -(g/\ell)\sin x_1 - (k/m)x_2 \end{bmatrix}, \quad h = x_1$$

Equilibrium Points:

$$\dot{x}_1 = \dot{x}_2 = 0 \quad \Rightarrow \quad x_2 = 0, \quad \sin x_1 = 0$$

$$\bar{x} = \begin{bmatrix} k\pi \\ 0 \end{bmatrix}, \quad k = 0, \pm 1, \pm 2, \ldots$$

$$f = \begin{bmatrix} x_2 \\ -(g/\ell)\sin x_1 - (k/m)x_2 \end{bmatrix}$$

$$\frac{\partial f}{\partial x} = \begin{bmatrix} 0 & 1 \\ -(g/\ell)\cos x_1 & -(k/m) \end{bmatrix}$$

$$\text{Linearization at } (0,0): \quad A = \begin{bmatrix} 0 & 1 \\ -(g/\ell) & -(k/m) \end{bmatrix}$$

$$\text{Linearization at } (\pi,0): \quad A = \begin{bmatrix} 0 & 1 \\ (g/\ell) & -(k/m) \end{bmatrix}$$

Classification of State models

Time-varying:

$$\begin{aligned} \dot{x} &= A(t)x + B(t)u \\ y &= C(t)x + D(t)u \end{aligned}$$

Time-invariant:

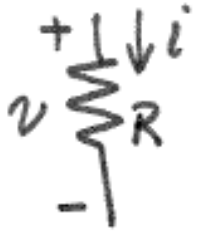
$$\begin{aligned} \dot{x} &= Ax + Bu \\ y &= Cx + Du \end{aligned}$$

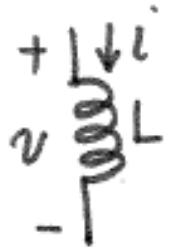
The response of a linear time-invariant system is invariant to shifts of the time origin. Therefore, without loss of generality, we can take the initial time to be $t_0 = 0$

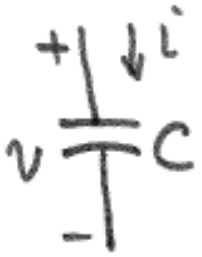
Examples of State models

RLC Circuits

Circuit Components

Resistor: 

Inductor: 

capacitor: 

Resistor: Ohm's Law

$$v(t) = R(t)i(t)$$

R is the resistance

Inductor:

$$v(t) = \dot{\psi}(t)$$

$\psi(t) = L(t)i(t)$ is magnetic flux through the inductor and L is the inductance. When L is time-invariant

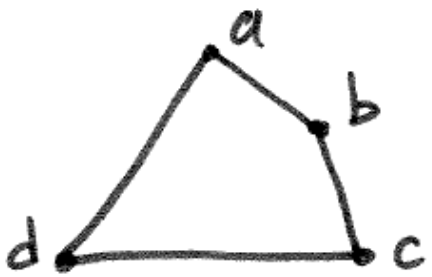
$$v(t) = L\frac{di}{dt}(t)$$

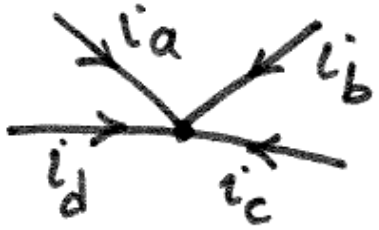
Capacitor:

$$i(t) = \dot{q}(t)$$

$q(t) = C(t)v(t)$ is the electric charge on the capacitor and C is the capacitance. When C is time invariant

$$i(t) = C\dot{v}(t)$$





Kirchoff's Voltage Law:

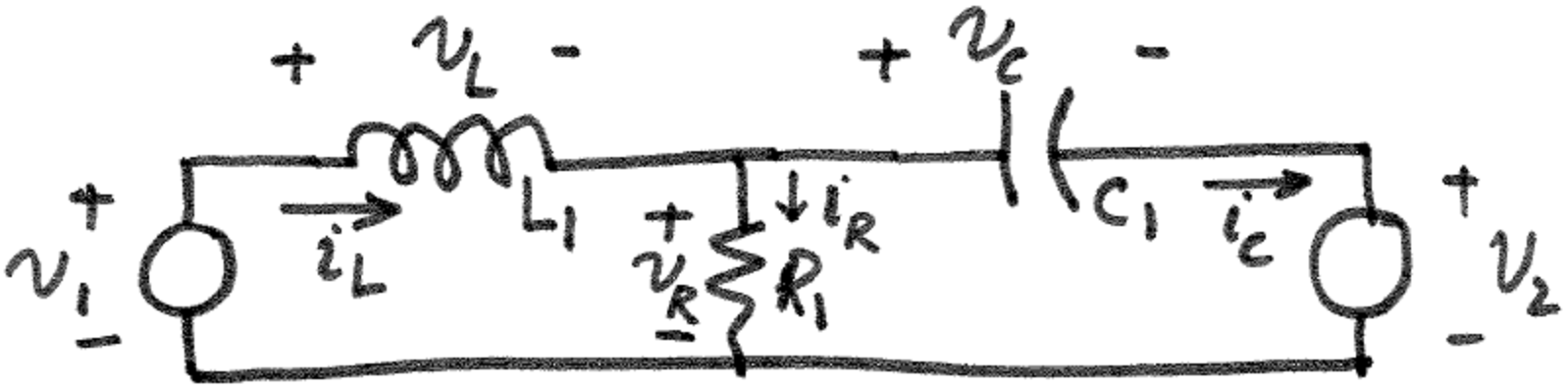
$$v_{ab} = v_a - v_b$$

$$v_{ab} + v_{bc} + v_{cd} + v_{da} = 0$$

Kirchoff's Current Law:

$$i_a + i_b + i_c + i_d = 0$$

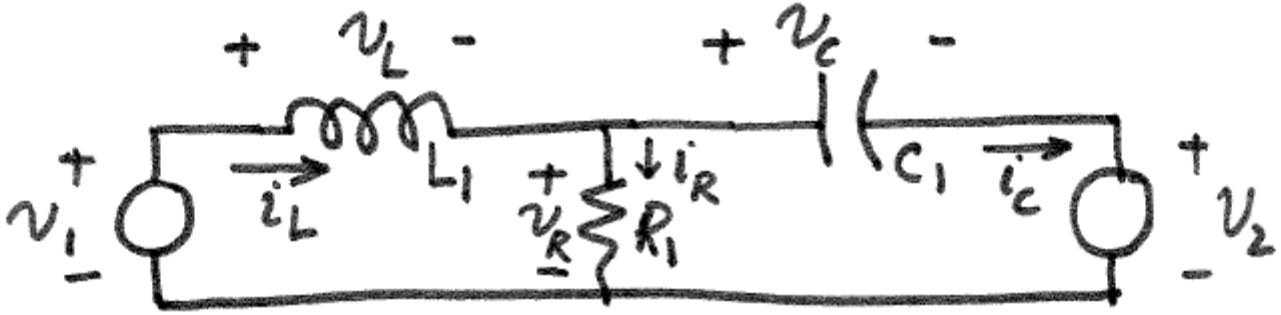
Example:



L_1 and C_1 could be time varying

$$u = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}, \quad y = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} i_L \\ i_C \end{bmatrix}$$

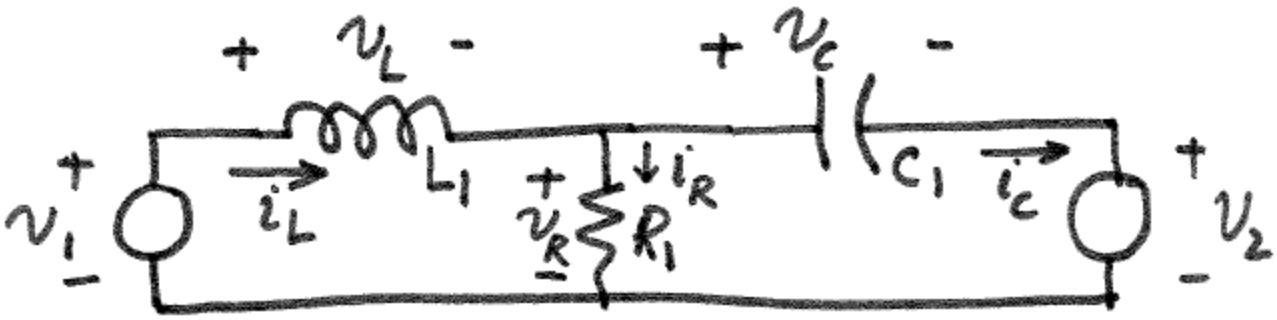
Case 1: Time-invariant L_1 and C_1



Choice of state variables:

$$x_1 = i_L, \quad x_2 = v_C$$

$$\begin{aligned} \dot{x}_1 &= \frac{di_L}{dt} = \frac{1}{L_1} v_L = \frac{1}{L_1}(v_1 - v_C - v_2) \\ &= -\frac{1}{L_1} x_2 + \frac{1}{L_1} u_1 - \frac{1}{L_1} u_2 \end{aligned}$$



$$
\begin{aligned}
\dot{x}_2 &= \frac{dv_C}{dt} = \frac{1}{C_1} i_C = \frac{1}{C_1}(i_L - i_R) \\
&= \frac{1}{C_1} i_L - \frac{1}{C_1 R_1} v_R = \frac{1}{C_1} i_L - \frac{1}{C_1 R_1}(v_C + v_2) \\
&= \frac{1}{C_1} x_1 - \frac{1}{C_1 R_1} x_2 - \frac{1}{C_1 R_1} u_2
\end{aligned}
$$

$$\dot{x}_1 = -\frac{1}{L_1}x_2 + \frac{1}{L_1}u_1 - \frac{1}{L_1}u_2$$

$$\dot{x}_2 = \frac{1}{C_1}x_1 - \frac{1}{C_1 R_1}x_2 - \frac{1}{C_1 R_1}u_2$$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} & \\ & \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} & \\ & \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$

$$\dot{x}_1 = -\frac{1}{L_1}x_2 + \frac{1}{L_1}u_1 - \frac{1}{L_1}u_2$$

$$\dot{x}_2 = \frac{1}{C_1}x_1 - \frac{1}{C_1 R_1}x_2 - \frac{1}{C_1 R_1}u_2$$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} & \\ & \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} & \\ & \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & -\frac{1}{L_1} \\ \frac{1}{C_1} & -\frac{1}{C_1 R_1} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} \frac{1}{L_1} & -\frac{1}{L_1} \\ 0 & -\frac{1}{C_1 R_1} \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$

$$\dot{x} = Ax + Bu$$

$$y_1 = i_L = x_1$$

$$y_2 = i_C = i_L - i_R = x_1 - \frac{1}{R_1}x_2 - \frac{1}{R_1}u_2$$

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} & \\ & \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} & \\ & \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$

$$y_1 = i_L = x_1$$

$$y_2 = i_C = i_L - i_R = x_1 - \frac{1}{R_1}x_2 - \frac{1}{R_1}u_2$$

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} & \\ & \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} & \\ & \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 1 & -\frac{1}{R_1} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & -\frac{1}{R_1} \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$

$$y = Cx + Du$$

Case 2: Time-varying L_1 and C_1

$$v_L = \frac{d}{dt}(L_1 i_L) = L_1 \frac{di_L}{dt} + \dot{L}_1 i_L$$

$$i_C = \frac{d}{dt}(C_1 v_C) = C_1 \dot{v}_C + \dot{C}_1 v_C$$

Choice of state variables:

$$x_1 = i_L, \quad x_2 = v_C$$

$$\dot{x}_1 = \frac{1}{L_1}(v_L - \dot{L}_1 i_L)$$

$$\dot{x}_2 = \frac{1}{C_1}(i_C - \dot{C}_1 v_C)$$

$$\dot{x}_1 = -\frac{\dot{L}_1}{L_1}x_1 - \frac{1}{L_1}x_2 + \frac{1}{L_1}u_1 - \frac{1}{L_1}u_2$$

$$\dot{x}_2 = \frac{1}{C_1}x_1 - \frac{\dot{C}_1}{C_1}x_2 - \frac{1}{C_1 R_1}x_2 - \frac{1}{C_1 R_1}u_2$$

$$\dot{x} = A(t)x + B(t)u$$

$$A = \begin{bmatrix} -\frac{\dot{L}_1}{L_1} & -\frac{1}{L_1} \\ \frac{1}{C_1} & -\frac{1}{C_1 R_1} - \frac{\dot{C}_1}{C_1} \end{bmatrix}, \quad B = \begin{bmatrix} \frac{1}{L_1} & -\frac{1}{L_1} \\ 0 & -\frac{1}{C_1 R_1} \end{bmatrix}$$

$$y = C(t)x + D(t)u$$

$$C = \begin{bmatrix} 1 & 0 \\ 1 & -\frac{1}{R_1} \end{bmatrix}, \quad D = \begin{bmatrix} 0 & 0 \\ 0 & -\frac{1}{R_1} \end{bmatrix}$$

Another choice of state variables:

$$x_1 = \psi = L_1 i_{\mathsf{L}}, \quad x_2 = q = C_1 v_{\mathsf{C}}$$

$$\dot{x}_1 = \dot{\psi} = v_{\mathsf{L}} = v_1 - v_{\mathsf{C}} - v_2 = -\frac{1}{C_1} x_2 + u_1 - u_2$$

$$\dot{x}_2 = \dot{q} = i_{\mathsf{C}} = i_{\mathsf{L}} - \frac{v_{\mathsf{C}} + v_2}{R_1} = \frac{1}{L_1} x_1 - \frac{1}{C_1 R_1} x_2 - \frac{1}{R_1} u_2$$

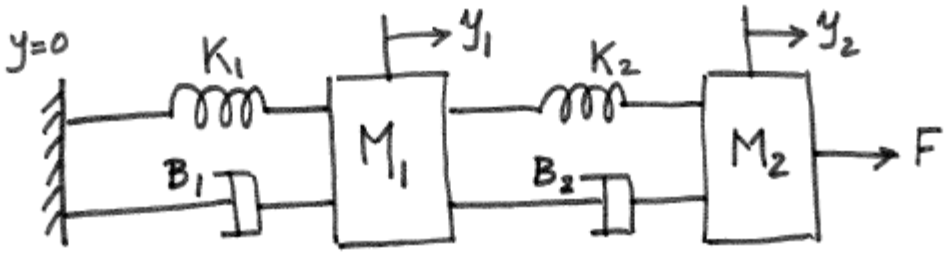
$$\dot{x} = A(t)x + B(t)u$$

$$A = \begin{bmatrix} 0 & -\frac{1}{C_1} \\ \frac{1}{L_1} & -\frac{1}{C_1 R_1} \end{bmatrix}, \quad B = \begin{bmatrix} 1 & -1 \\ 0 & -\frac{1}{R_1} \end{bmatrix}$$

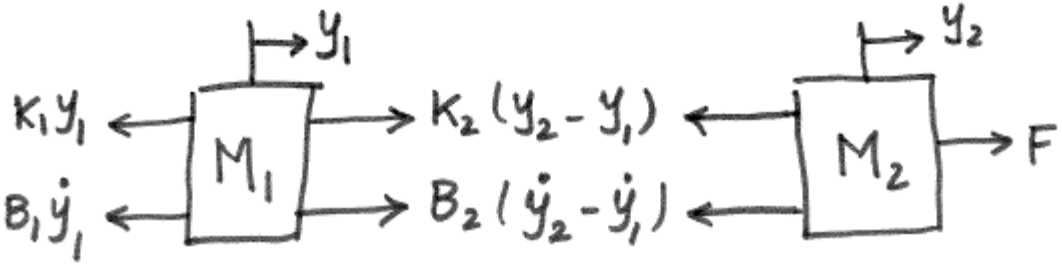
State models are not unique

Mechanical Systems

Example:



Free-body diagrams



Newton's Law

$$M_1\ddot{y}_1 = K_2(y_2 - y_1) + B_2(\dot{y}_2 - \dot{y}_1) - K_1 y_1 - B_1\dot{y}_1$$

$$M_2\ddot{y}_2 = F - K_2(y_2 - y_1) - B_2(\dot{y}_2 - \dot{y}_1)$$

$$M_1\ddot{y}_1 = K_2(y_2 - y_1) + B_2(\dot{y}_2 - \dot{y}_1) - K_1 y_1 - B_1\dot{y}_1$$

$$M_2\ddot{y}_2 = F - K_2(y_2 - y_1) - B_2(\dot{y}_2 - \dot{y}_1)$$

Choice of state variables:

$$x_1 = y_1, \quad x_2 = y_2, \quad x_3 = \dot{y}_1, \quad x_4 = \dot{y}_2$$

$$\dot{x}_1 = \dot{y}_1 = x_3$$

$$\dot{x}_2 = \dot{y}_2 = x_4$$

$$M_1\ddot{y}_1 = K_2(y_2 - y_1) + B_2(\dot{y}_2 - \dot{y}_1) - K_1 y_1 - B_1\dot{y}_1$$

$$M_2\ddot{y}_2 = F - K_2(y_2 - y_1) - B_2(\dot{y}_2 - \dot{y}_1)$$

$$\begin{aligned}
\dot{x}_3 &= \ddot{y}_1 \\
&= \frac{K_2}{M_1}(y_2 - y_1) + \frac{B_2}{M_1}(\dot{y}_2 - \dot{y}_1) - \frac{K_1}{M_1}y_1 - \frac{B_1}{M_1}\dot{y}_1 \\
&= -\left(\frac{K_1 + K_2}{M_1}\right)x_1 + \frac{K_2}{M_1}x_2 - \left(\frac{B_1 + B_2}{M_1}\right)x_3 \\
&\quad + \frac{B_2}{M_1}x_4
\end{aligned}$$

$$M_1\ddot{y}_1 = K_2(y_2 - y_1) + B_2(\dot{y}_2 - \dot{y}_1) - K_1 y_1 - B_1\dot{y}_1$$

$$M_2\ddot{y}_2 = F - K_2(y_2 - y_1) - B_2(\dot{y}_2 - \dot{y}_1)$$

$$\begin{aligned}
\dot{x}_4 &= \ddot{y}_2 \\
&= \frac{1}{M_2}F - \frac{K_2}{M_2}(y_2 - y_1) - \frac{B_2}{M_2}(\dot{y}_2 - \dot{y}_1) \\
&= \frac{K_2}{M_2}x_1 - \frac{K_2}{M_2}x_2 + \frac{B_2}{M_2}x_3 - \frac{B_2}{M_2}x_4 + \frac{1}{M_2}u
\end{aligned}$$

$$\begin{aligned}
y_1 &= x_1 \\
y_2 &= x_2
\end{aligned}$$

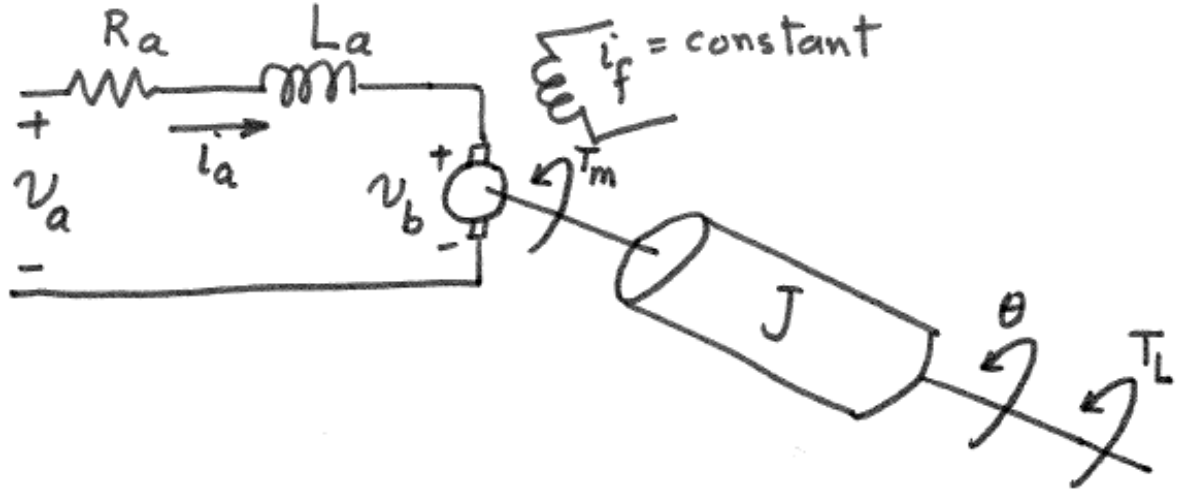
$$\begin{aligned} \dot{x} &= Ax + Bu \\ y &= Cx + Du \end{aligned}$$

$$A = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -\left(\frac{K_1+K_2}{M_1}\right) & \frac{K_2}{M_1} & -\left(\frac{B_1+B_2}{M_1}\right) & \frac{B_2}{M_1} \\ \frac{K_2}{M_2} & \frac{-K_2}{M_2} & \frac{B_2}{M_2} & \frac{-B_2}{M_2} \end{bmatrix}, B = \begin{bmatrix} 0 \\ 0 \\ 0 \\ \frac{1}{M_2} \end{bmatrix}$$

$$C = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}, D = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Electromechanical Systems

Example: Armature Controlled DC Motor



$$T_m = K_m i_a, \quad v_b = K_b \omega = K_b \dot{\theta}$$

$$v_a = i_a R_a + L_a \frac{di_a}{dt} + K_b \dot{\theta}$$

$$J\ddot{\theta} = K_m i_a + T_L - D\dot{\theta}$$

Inputs: $u = v_a, \quad d = T_L$

Output: For position control $y = \theta$

Choice of state variables:

$$x_1 = \theta, \quad x_2 = \dot{\theta}, \quad x_3 = i_a$$

$$\begin{aligned}
\dot{x}_1 &= x_2 \\
\dot{x}_2 &= (K_m i_a + T_L - D\dot{\theta})/J \\
&= -(D/J)x_2 + (K_m/J)x_3 + (1/J)d \\
\dot{x}_3 &= \frac{di_a}{dt} = (v_a - R_a i_a - K_b\dot{\theta})/L_a \\
&= -(K_b/L_a)x_2 - (R_a/L_a)x_3 + (1/L_a)u \\
y &= x_1
\end{aligned}$$

$$\begin{aligned} \dot{x} &= Ax + Bu + Ed \\ y &= Cx \end{aligned}$$

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & -(D/J) & (K_m/J) \\ 0 & -(K_b/L_a) & -(R_a/L_a) \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 0 \\ 1/L_a \end{bmatrix}$$

$$E = \begin{bmatrix} 0 \\ 1/J \\ 0 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}$$

Consider two cases:

Case 1: load torque comes from viscous damper

$$T_L = -b\dot{\theta} \quad \Rightarrow \quad Ed = -b \begin{bmatrix} 0 \\ 1/J \\ 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \end{bmatrix} x$$

Case 2: load torque comes from rotational spring

$$T_L = -k\theta \quad \Rightarrow \quad Ed = -k \begin{bmatrix} 0 \\ 1/J \\ 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} x$$

Case 1: $\dot{x} = Ax + Bu, \quad y = Cx$

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & -(D+b)/J & (K_m/J) \\ 0 & -(K_b/L_a) & -(R_a/L_a) \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 0 \\ 1/L_a \end{bmatrix}$$

$$C = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}$$

Case 2: $\dot{x} = Ax + Bu, \quad y = Cx$

$$A = \begin{bmatrix} 0 & 1 & 0 \\ -(k/J) & -(D/J) & (K_m/J) \\ 0 & -(K_b/L_a) & -(R_a/L_a) \end{bmatrix}$$

Summarize

State: Definition

The state of a dynamical system is the smallest set of variables such that knowledge of these variables at time t_0 together with knowledge of the input for all time $t \geq t_0$, completely describes the behavior of the system for all time $t \geq t_0$.

Number of state variables is equal to number of initial conditions required to solve the ordinary differential equations.

In the state-space approach, we work with first-order differential equations. Therefore, the number of states is equal to the number of differential equations.

Mathematical Preliminaries

Vector and Matrix Norms

For an n-dimensional vector x ($x \in R^n$)

$$\|x\| = \sqrt{x_1^2 + \cdots + x_n^2} = \sqrt{x^T x}$$

Properties:

- $\|x\| \geq 0$ for all $x \in R^n$
- $\|x\| = 0$ if and only if $x = 0$
- $\|x + y\| \leq \|x\| + \|y\|$ for all $x, y \in R^n$
- $\|\alpha x\| = |\alpha| \|x\|$ for all $\alpha \in R$ and $x \in R^n$

For an $m \times n$ matrix A

$$\|A\| = \max_{\|x\|=1} \|Ax\| = \sqrt{\lambda_{max}(A^T A)}$$

Properties:

- $\|Ax\| \leq \|A\| \, \|x\|$
- $\|AB\| \leq \|A\| \, \|B\|$
- $\|A + B\| \leq \|A\| + \|B\|$

Nonsingular Matrix

A square matrix A is nonsingular if $\det(A) \neq 0$

If A, B, C are square matrices of the same dimension,

$$C = AB$$

and C is nonsingular, then both A and B are nonsingular because

$$\det(C) = \det(A)\det(B)$$

If I is the identity matrix and

$$I = AB$$

then B is called the inverse of A (denoted by A^{-1})

Also $A = B^{-1}$

Convergence

For an infinite series of vector functions

$$\sum_{j=0}^{\infty} f_j(t)$$

where $f_j(t)$ is defined for $t \in [t_0, t_1]$, if

- $\|f_j(t)\| \leq \alpha_j$ for all $t \in [t_0, t_1]$ and every j, and
- the series $\sum_{j=1}^{\infty} \alpha_j$ is convergent

then the series $\sum_{j=0}^{\infty} f_j(t)$ is convergent on $[t_0, t_1]$

Solution of $\dot{x} = Ax$

$$\dot{x} = Ax, \quad x(0) = x_0$$

$$x(t) = x_0 + \int_0^t Ax(\tau)\, d\tau = x_0 + A\int_0^t x(\tau)\, d\tau$$

Successive Approximation:

$$x_0(t) = x_0$$

$$x_1(t) = x_0 + A\int_0^t x_0(\tau)\, d\tau = x_0 + tAx_0$$

$$x_1(t) = x_0 + tAx_0$$

$$x_2(t) = x_0 + A\int_0^t x_1(\tau)\,d\tau = x_0 + tAx_0 + \frac{t^2}{2}A^2x_0$$

$$x_3(t) = x_0 + A\int_0^t x_2(\tau)\,d\tau = x_0 + tAx_0 + \frac{t^2}{2}A^2x_0 + \frac{t^3}{6}A^3x_0$$

$$x_k(t) = \sum_{j=0}^{k} \frac{t^j}{j!}A^jx_0 \qquad [A^0 = I]$$

Will the series converge as $k \to \infty$?

$$x_k(t) = \sum_{j=0}^{k} \frac{t^j}{j!} A^j x_0$$

$$\left\| \frac{t^j}{j!} A^j x_0 \right\| \leq \frac{|t|^j}{j!} \|A\|^j \|x_0\| \leq \frac{(T\|A\|)^j}{j!} \|x_0\|$$

for all $t \in [-T, T]$. The series

$$\sum_{j=0}^{\infty} \frac{(T\|A\|)^j}{j!} \|x_0\| = e^{T\|A\|} \|x_0\|$$

is convergent. Hence, the series $\sum_{j=0}^{\infty} \frac{t^j}{j!} A^j x_0$ is convergent

Is

$$\sum_{j=0}^{\infty} \frac{t^j}{j!} A^j x_0$$

the solution of

$$\dot{x} = Ax, \quad x(0) = x_0$$

$$\sum_{j=0}^{\infty} \frac{t^j}{j!} A^j x_0 \Big|_{t=0} = A^0 x_0 = x_0$$

$$
\begin{aligned}
&\frac{d}{dt}\sum_{j=0}^{\infty}\frac{t^j}{j!}A^j x_0 \\
&= \frac{d}{dt}\left(x_0 + tAx_0 + \frac{t^2}{2!}A^2x_0 + \frac{t^3}{6!}A^3x_0 + \cdots\right) \\
&= Ax_0 + tA^2x_0 + \frac{t^2}{2!}A^3x_0 + \cdots \\
&= A\left(x_0 + tAx_0 + \frac{t^2}{2!}A^2x_0 + \cdots\right) \\
&= A\sum_{j=0}^{\infty}\frac{t^j}{j!}A^j x_0
\end{aligned}
$$

Definition: For a square matrix M, the exponential matrix e^M (also written $\exp(M)$) is defined by the convergent infinite series

$$e^M = \sum_{j=0}^{\infty} \frac{1}{j!} M^j$$

The solution of

$$\dot{x} = Ax, \quad x(0) = x_0$$

is given by

$$x(t) = e^{At} x_0$$

$$e^{At} = \sum_{j=0}^{\infty} \frac{t^j}{j!} A^j$$

Properties of e^{At}:

- $e^0 = I$
- $\frac{d}{dt}e^{At} = Ae^{At}$

Because

$$\frac{d}{dt}e^{At}x_0 = Ae^{At}x_0$$

is valid for all x_0

Complete Solution

The solution of

$$\dot{x} = Ax + Bu, \quad x(0) = x_0$$

is given by

$$x(t) = e^{At}x_0 + \int_0^t e^{A(t-\tau)}Bu(\tau)\,d\tau$$

Proof:

$$x(0) = e^0 x_0 + \int_0^0 e^{A(-\tau)}Bu(\tau)\,d\tau = x_0$$

$$
\begin{aligned}
\dot{x} &= \frac{d}{dt}\left(e^{At}x_0 + \int_0^t e^{A(t-\tau)}Bu(\tau)\,d\tau\right) \\
&= Ae^{At}x_0 + \int_0^t Ae^{A(t-\tau)}Bu(\tau)\,d\tau \\
&\quad + e^{A(t-\tau)}Bu(\tau)\Big|_{\tau=t} \\
&= A\left(e^{At}x_0 + \int_0^t e^{A(t-\tau)}Bu(\tau)\,d\tau\right) + Bu(t) \\
&= Ax + Bu
\end{aligned}
$$

Matrix Exponential

Example: Find e^{At} for

$$A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$A^k = 0, \quad \forall\, k \geq 2$$

$$e^{At} = \sum_{j=0}^{\infty} \frac{t^j}{j!} A^j = I + tA = \begin{bmatrix} 1 & t \\ 0 & 1 \end{bmatrix}$$

Example: Find e^{At} for

$$A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} = -I$$

$$A^3 = A^2 A = -A$$

$$A^4 = A^3 A = (-A)(A) = -A^2 = I$$

$$A^5 = A^4 A = A$$

$$A^k = \begin{cases} (-1)^{(k-1)/2} A, & k = 1, 3, 5, \ldots \\ \\ (-1)^{k/2} I, & k = 0, 2, 4, 6, \ldots \end{cases}$$

$$\begin{aligned} e^{At} &= \sum_{j=0}^{\infty} \frac{t^j}{j!} A^j \\ &= \left(1 - \frac{t^2}{2!} + \frac{t^4}{4!} + \cdots\right) I \\ &\quad + \left(t - \frac{t^3}{3!} + \frac{t^5}{5!} - \cdots\right) A \\ &= \cos t \; I + \sin t \; A \end{aligned}$$

$$
\begin{aligned}
e^{At} &= \cos t\, I + \sin t\, A \\
&= \cos t \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \sin t \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \\
&= \begin{bmatrix} \cos t & \sin t \\ -\sin t & \cos t \end{bmatrix}
\end{aligned}
$$

Properties of e^{At}:

- $e^0 = I$
- $\frac{d}{dt}e^{At} = Ae^{At} = e^{At}A$
- $e^{A(t+\tau)} = e^{At}e^{A\tau}$
- e^{At} is always nonsingular and $\left(e^{At}\right)^{-1} = e^{-At}$
- $e^{A+B} = e^{A}e^{B}$ if and only if $AB = BA$

Why $\frac{d}{dt}e^{At} = e^{At}A$?

$$\begin{aligned}\frac{d}{dt}e^{At} &= A + tA^2 + \frac{t^2}{2!}A^3 + \frac{t^3}{3!}A^4 + \cdots \\ &= \left(I + tA + \frac{t^2}{2!}A^2 + \frac{t^3}{3!}A^3 + \cdots\right)A\end{aligned}$$

Why $e^{A(t+\tau)} = e^{At}e^{A\tau}$?

$$x(t+\tau) = e^{At}x(\tau) = e^{At}e^{A\tau}x(0)$$

$$x(t+\tau) = e^{A(t+\tau)}x(0)$$

By the uniqueness of solutions

$$e^{A(t+\tau)}x(0) = e^{At}e^{A\tau}x(0)$$

Since this identity is valid for all $x(0)$

$$e^{A(t+\tau)} = e^{At}e^{A\tau}$$

Why $\left(e^{At}\right)^{-1} = e^{-At}$?

Set $\tau = -t$ in

$$e^{A(t+\tau)} = e^{At}e^{A\tau}$$

to obtain

$$I = e^{At}e^{-At}$$

$$\Rightarrow e^{-At} = \left(e^{At}\right)^{-1}$$

Why $e^{A+B} = e^A e^B$ if and only if $AB = BA$?

$$
\begin{aligned}
e^{A+B} &= I + (A+B) + \frac{1}{2!}(A+B)(A+B) + \cdots \\
&= I + A + B + \frac{1}{2!}(A^2 + AB + BA + B^2) + \cdots
\end{aligned}
$$

$$
\begin{aligned}
e^A\, e^B &= \left(I + A + \frac{1}{2!}A^2 + \cdots\right)\left(I + B + \frac{1}{2!}B^2 + \cdots\right) \\
&= I + A + B + \frac{1}{2!}(A^2 + 2AB + B^2) + \cdots
\end{aligned}
$$

Example:

$$A = \lambda I + B$$

$$e^{At} = e^{(\lambda I + B)t} = e^{\lambda t I} e^{Bt} = e^{\lambda t} e^{Bt}$$

Example: Find e^{At} for

$$A = \begin{bmatrix} -\alpha & 1 \\ -1 & -\alpha \end{bmatrix}$$

$$A = -\alpha I + B, \quad \text{where } B = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$$

$$e^{At} = e^{-\alpha t} e^{Bt} = e^{-\alpha t} \begin{bmatrix} \cos t & \sin t \\ -\sin t & \cos t \end{bmatrix}$$

Example: Find e^{At} for

$$A = \begin{bmatrix} \lambda & 1 & 0 \\ 0 & \lambda & 1 \\ 0 & 0 & \lambda \end{bmatrix}$$

$$A = \lambda I + B, \quad \text{where } B = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$B^2 = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \quad B^k = 0 \text{ for } k \geq 3$$

$$e^{Bt} = \begin{bmatrix} 1 & t & \frac{1}{2!}t^2 \\ 0 & 1 & t \\ 0 & 0 & 1 \end{bmatrix}$$

$$e^{At} = e^{\lambda t}e^{Bt} = e^{\lambda t}\begin{bmatrix} 1 & t & \frac{1}{2!}t^2 \\ 0 & 1 & t \\ 0 & 0 & 1 \end{bmatrix}$$

For

$$A = \begin{bmatrix} \lambda & 1 & 0 & 0 \\ 0 & \lambda & 1 & 0 \\ 0 & 0 & \lambda & 1 \\ 0 & 0 & 0 & \lambda \end{bmatrix}$$

$$e^{At} = e^{\lambda t} \begin{bmatrix} 1 & t & \frac{1}{2!}t^2 & \frac{1}{3!}t^3 \\ 0 & 1 & t & \frac{1}{2!}t^2 \\ 0 & 0 & 1 & t \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Laplace Transform Method

Review of Laplace Transform

The Laplace transform of $f(t)$ is defined by

$$\hat{f}(s) = \mathcal{L}\{f(t)\} = \int_0^\infty f(t)e^{-st}\,dt$$

Properties

$$\mathcal{L}\left\{\frac{df(t)}{dt}\right\} = s\hat{f}(s) - f(0)$$

$$\mathcal{L}\left\{\frac{d^k f(t)}{dt^k}\right\} = s^k\hat{f}(s) - [s^{k-1}f(0) + \cdots + f^{(k-1)}(0)]$$

$$\mathcal{L}\left\{e^{-at}f(t)\right\} = \hat{f}(s+a)$$

$$\mathcal{L}\{f(t-a)p(t-a)\} = e^{-as}\hat{f}(s), \quad a > 0$$

$$\mathcal{L}\{f(t/\alpha)\} = \alpha\hat{f}(\alpha s), \quad \alpha > 0$$

$$\mathcal{L}\left\{\int_0^t f(\tau)g(t-\tau)\,d\tau\right\} = \hat{f}(s)\hat{g}(s)$$

$$f(0^+) = \lim_{t\to 0^+} f(t) = \lim_{s\to\infty} s\hat{f}(s)$$

provided the limit exists

$$\lim_{t\to\infty} f(t) = \lim_{s\to 0} s\hat{f}(s)$$

provided all poles of $s\hat{f}(s)$ have negative real parts

Solution of $\dot{x} = Ax$

$$\dot{x}(t) = Ax(t), \quad x(0) = x_0$$

$$s\hat{x}(s) - x(0) = A\hat{x}(s)$$

$$s\hat{x}(s) - A\hat{x}(s) = x_0$$

$$sI\hat{x}(s) - A\hat{x}(s) = x_0$$

$$(sI - A)\hat{x}(s) = x_0$$

$$\hat{x}(s) = (sI - A)^{-1}x_0$$

$$x(t) = \mathcal{L}^{-1}\left\{(sI - A)^{-1}x_0\right\} = \mathcal{L}^{-1}\left\{(sI - A)^{-1}\right\}x_0$$

$$x(t) = \mathcal{L}^{-1}\left\{(sI - A)^{-1}\right\} x_0$$

But

$$x(t) = e^{At} x_0$$

By the uniqueness of solutions

$$e^{At} = \mathcal{L}^{-1}\left\{(sI - A)^{-1}\right\}$$

Example: Find e^{At} for

$$A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$$

$$(sI - A)^{-1} = \begin{bmatrix} s & -1 \\ 1 & s \end{bmatrix}^{-1} = \frac{1}{s^2+1} \begin{bmatrix} s & 1 \\ -1 & s \end{bmatrix}$$

$$(sI - A)^{-1} = \begin{bmatrix} \frac{s}{s^2+1} & \frac{1}{s^2+1} \\ -\frac{1}{s^2+1} & \frac{s}{s^2+1} \end{bmatrix}$$

$$e^{At} = \begin{bmatrix} \cos t & \sin t \\ -\sin t & \cos t \end{bmatrix}$$

Example: Find e^{At} for

$$A = \begin{bmatrix} -1 & 1 \\ -\frac{1}{2} & -2 \end{bmatrix}$$

$$(sI - A)^{-1} = \begin{bmatrix} s+1 & -1 \\ \frac{1}{2} & s+2 \end{bmatrix}^{-1}$$

$$\det(sI-A) = (s+1)(s+2)+\frac{1}{2} = s^2+3s+\frac{5}{2} = \left(s+\frac{3}{2}\right)^2+\frac{1}{4}$$

$$(sI - A)^{-1} = \frac{1}{\left(s+\frac{3}{2}\right)^2+\frac{1}{4}} \begin{bmatrix} s+2 & 1 \\ -\frac{1}{2} & s+1 \end{bmatrix}$$

$$(sI-A)^{-1} = \begin{bmatrix} \frac{s+\frac{3}{2}}{\left(s+\frac{3}{2}\right)^2+\frac{1}{4}} + \frac{\frac{1}{2}}{\left(s+\frac{3}{2}\right)^2+\frac{1}{4}} & \frac{2\cdot\frac{1}{2}}{\left(s+\frac{3}{2}\right)^2+\frac{1}{4}} \\ \frac{-\frac{1}{2}}{\left(s+\frac{3}{2}\right)^2+\frac{1}{4}} & \frac{s+\frac{3}{2}}{\left(s+\frac{3}{2}\right)^2+\frac{1}{4}} - \frac{\frac{1}{2}}{\left(s+\frac{3}{2}\right)^2+\frac{1}{4}} \end{bmatrix}$$

$$e^{At} = e^{-\frac{3t}{2}} \begin{bmatrix} \cos\left(\frac{t}{2}\right) + \sin\left(\frac{t}{2}\right) & 2\sin\left(\frac{t}{2}\right) \\ -\sin\left(\frac{t}{2}\right) & \cos\left(\frac{t}{2}\right) - \sin\left(\frac{t}{2}\right) \end{bmatrix}$$

Solution of $\dot{x} = Ax + Bu$

$$\dot{x}(t) = Ax(t) + Bu(t), \qquad x(0) = x_0$$

$$s\hat{x}(s) - x(0) = A\hat{x}(s) + B\hat{u}(s)$$

$$s\hat{x}(s) - A\hat{x}(s) = x_0 + B\hat{u}(s)$$

$$(sI - A)\hat{x}(s) = x_0 + B\hat{u}(s)$$

$$\hat{x}(s) = (sI - A)^{-1}x_0 + (sI - A)^{-1}B\hat{u}(s)$$

$$x(t) = \mathcal{L}^{-1}\left\{(sI - A)^{-1}\right\}x_0 + \mathcal{L}^{-1}\left\{(sI - A)^{-1}B\hat{u}(s)\right\}$$

$$x(t) = \mathcal{L}^{-1}\left\{(sI - A)^{-1}\right\} x_0 + \mathcal{L}^{-1}\left\{(sI - A)^{-1} B\hat{u}(s)\right\}$$

But

$$\mathcal{L}^{-1}\left\{(sI - A)^{-1}\right\} = e^{At}$$

Therefore

$$x(t) = e^{At}x_0 + \int_0^t e^{A(t-\tau)} Bu(\tau)\, d\tau$$

Input-Output Response

$$\begin{aligned} \dot{x} &= Ax + Bu \\ y &= Cx + Du \end{aligned}$$

Take Laplace transform of the state equation

$$s\hat{x}(s) - x(0) = A\hat{x}(s) + B\hat{u}(s)$$

Assume $x(0) = 0$

$$(sI - A)\hat{x}(s) = B\hat{u}(s)$$

$$\hat{x}(s) = (sI - A)^{-1}B\hat{u}(s)$$

Take Laplace transform of the output equation

$$\hat{y}(s) = C\hat{x}(s) + D\hat{u}(s)$$

$$\begin{aligned} \hat{y}(s) &= C(sI - A)^{-1}B\hat{u}(s) + D\hat{u}(s) \\ &= [C(sI - A)^{-1}B + D]\hat{u}(s) \end{aligned}$$

Transfer Function: The transfer function $H(s)$ is defined by

$$\hat{y}(s) = H(s)\hat{u}(s)$$

($H_{ij}(s)$ is the transfer function from u_j to y_i)

$$H(s) = C(sI - A)^{-1}B + D$$

State Transformations

$$\begin{aligned} \dot{x} &= Ax + Bu \\ y &= Cx + Du \end{aligned}$$

Change of variables:

$$x(t) = Pz(t)$$

P is a nonsingular matrix

$$z(t) = P^{-1}x(t)$$

$$\dot{z} = P^{-1}\dot{x} = P^{-1}(Ax + Bu) = P^{-1}APz + P^{-1}Bu$$

$$y = Cx + Du = CPz + Du$$

$$\{A, B, C, D\} \rightarrow \boxed{x = Pz} \rightarrow \{P^{-1}AP, P^{-1}B, CP, D\}$$

What is the effect on the exponential matrix?

$$\dot{x} = Ax \quad \Rightarrow \quad x(t) = e^{At}x(0)$$

$$\dot{z} = P^{-1}APz \quad \Rightarrow \quad z(t) = e^{P^{-1}APt}z(0)$$

$$x(t) = Pz(t) \;\; \& \;\; z(0) = P^{-1}x(0)$$

$$x(t) = Pe^{P^{-1}APt}P^{-1}x(0)$$

By the uniqueness of solutions

$$e^{At}x(0) = Pe^{P^{-1}APt}P^{-1}x(0)$$

$$\Rightarrow e^{At} = Pe^{P^{-1}APt}P^{-1}$$

What is the effect on the transfer function?

The transfer function in the original coordinates:

$$H(s) = C(sI - A)^{-1}B + D$$

The transfer function in the transformed coordinates:

$$\begin{aligned}
\tilde{H}(s) &= CP\left(sI - P^{-1}AP\right)^{-1}P^{-1}B + D \\
&= CP\left(sP^{-1}IP - P^{-1}AP\right)^{-1}P^{-1}B + D \\
&= CP\left[P^{-1}\left(sI - A\right)P\right]^{-1}P^{-1}B + D \\
&= CPP^{-1}(sI - A)^{-1}PP^{-1}B + D \\
&= C(sI - A)^{-1}B + D \;=\; H(s)
\end{aligned}$$

The transfer function is invariant under state transformations

Thank You